

BS COMPUTER SCIENCE · 3RD SEMESTER

ARTIFICIAL INTELLIGENCE

Lecture 01 — Introduction to Artificial Intelligence

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What Comes to Your Mind When You Hear "AI"?



ChatGPT



Self-driving cars



Face recognition



Recommendations



Robots



Voice assistants



Image generation



Fraud detection

"Write ONE example of AI that YOU used today."

Think for 60 seconds, then we will ask 3–5 students to share. (Don't reveal the answer yet.)

What Is Artificial Intelligence?

The field of computing concerned with building systems that can perform tasks requiring capabilities commonly associated with human intelligence.



● Learning

● Reasoning

● Problem solving

● Perception

● Language understanding

● Decision making

● Planning

● Generation

Human Intelligence vs. Artificial Intelligence



Human Intelligence

- Learns from lived experience
- Uses common sense
- Understands deep context
- Adapts to totally new situations
- Reasons flexibly
- Communicates naturally



Artificial Intelligence

- Learns from data
- Detects statistical patterns
- Optimizes decisions at scale
- Generates new content
- Performs specific, trained tasks
- Processes huge volumes of information

Most AI systems today are specialized / task-oriented — not equivalent to general human intelligence.

Is This AI?



Calculator

AI? Not AI? Depends?



Spam filter

AI? Not AI? Depends?



Face recognition

AI? Not AI? Depends?



Maps route planning

AI? Not AI? Depends?



Netflix recommendation

AI? Not AI? Depends?



ChatGPT

AI? Not AI? Depends?



Thermostat

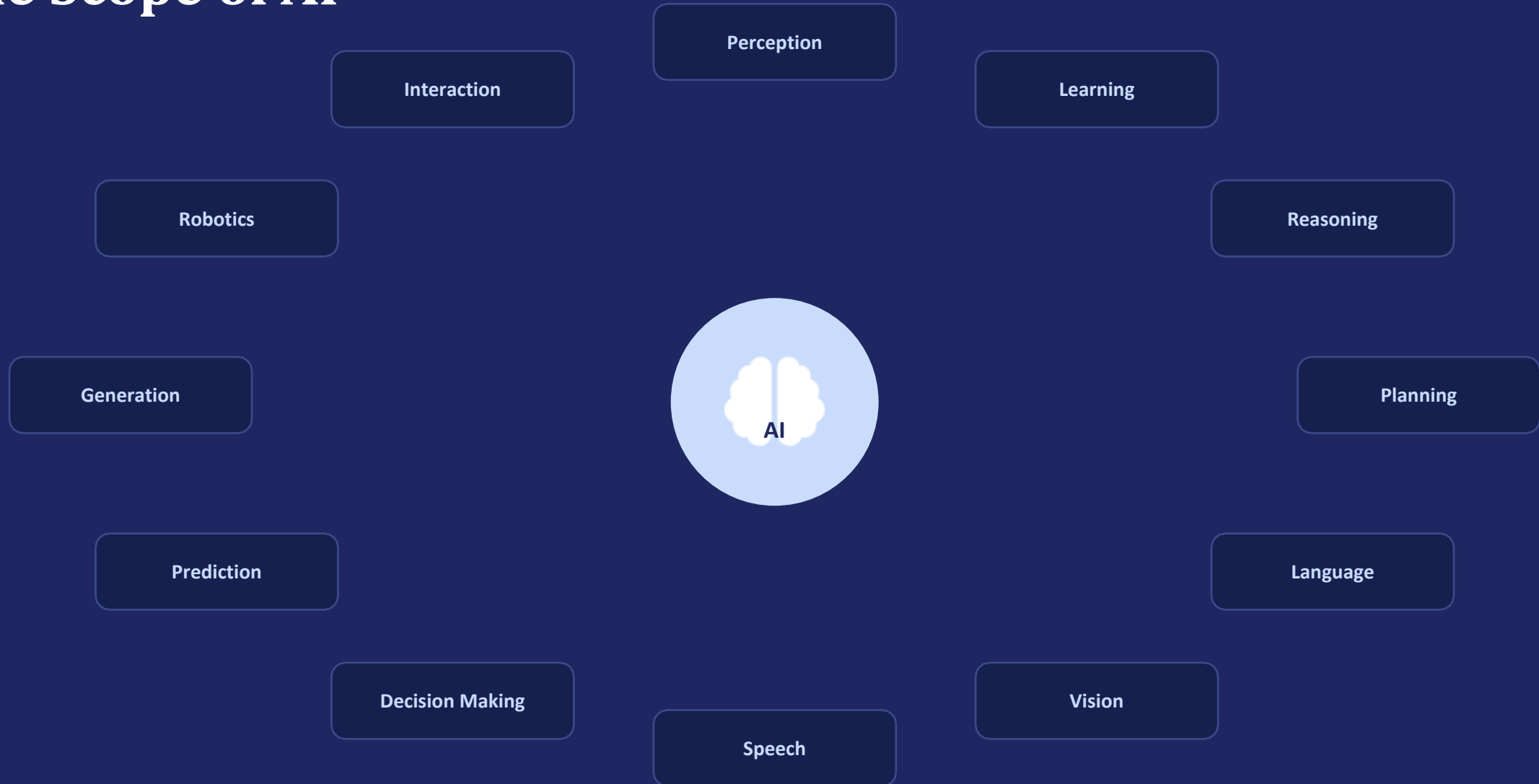
AI? Not AI? Depends?



Autonomous vehicle

AI? Not AI? Depends?

The Scope of AI

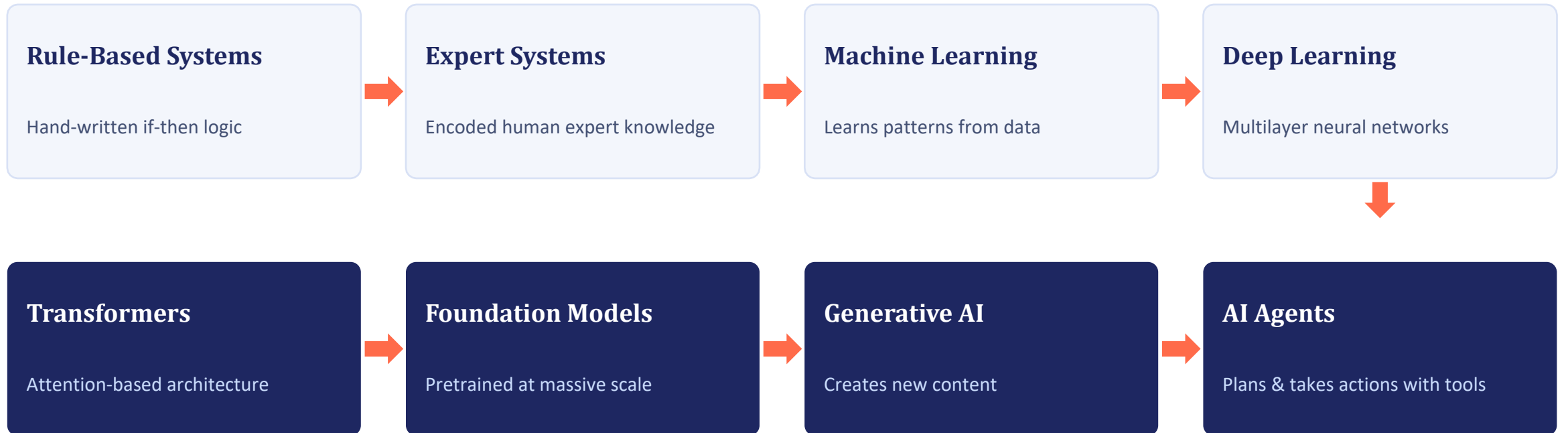


A Brief History of AI

1940s: early neural / computational ideas precede this timeline.



From Rules to Generative AI



AI vs. ML vs. DL vs. Generative AI

Artificial Intelligence

the broad field of intelligent computational systems

Machine Learning

systems learn patterns from data

Deep Learning

ML using multilayer neural networks

Generative AI

creates new content — commonly built on deep learning & foundation models

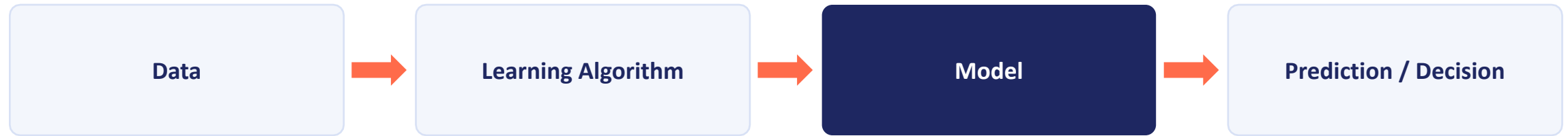


Important nuance

Generative AI is NOT always a strict subset of Deep Learning in every possible context — but modern Generative AI systems commonly rely on deep learning and large foundation models.

"Which layer does a self-driving car's object detector belong to? What about a chatbot that writes essays?"

Machine Learning



Example: Email → [Learning Algorithm] → Spam / Not Spam

Supervised Learning

Learns from labeled examples (input → known correct output)

Unsupervised Learning

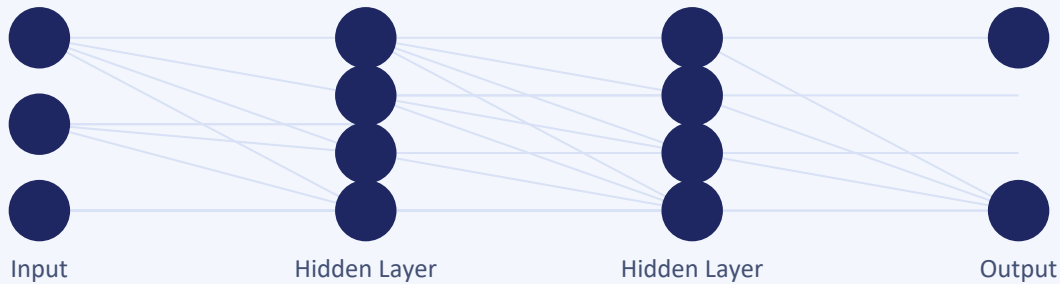
Finds structure/patterns in unlabeled data

Reinforcement Learning

Learns by trial, error, and reward feedback

Deep Learning

Neural networks with multiple layers that learn useful representations from data.



Examples

- Image recognition
- Speech recognition
- Language understanding

CNN

Convolutional Neural Network — strong for images

RNN

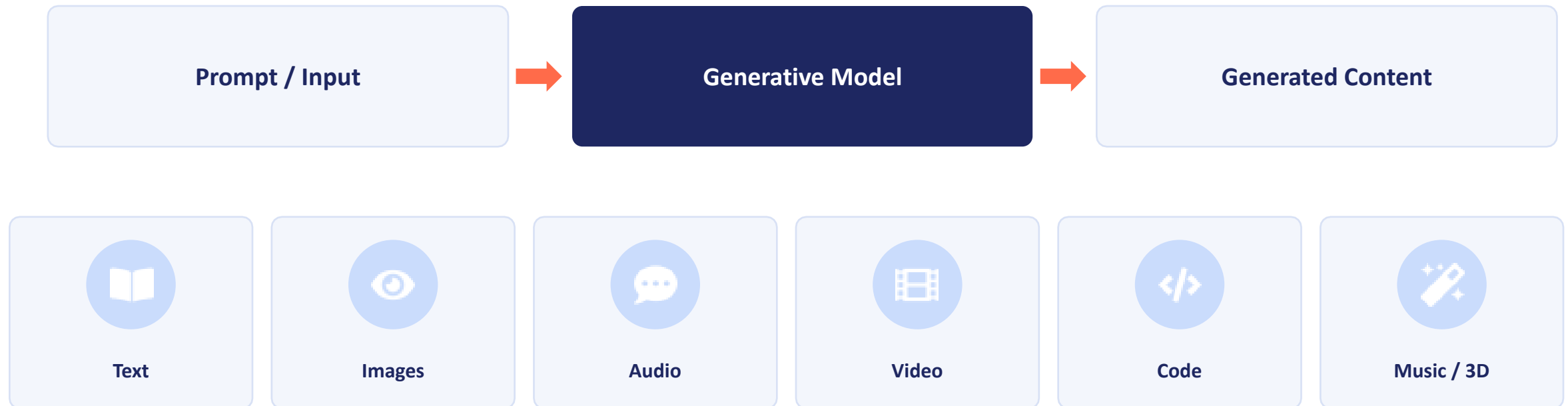
Recurrent Neural Network — strong for sequences

Transformer

Attention-based — powers modern LLMs

(We will not go deep into these architectures today — just names for now.)

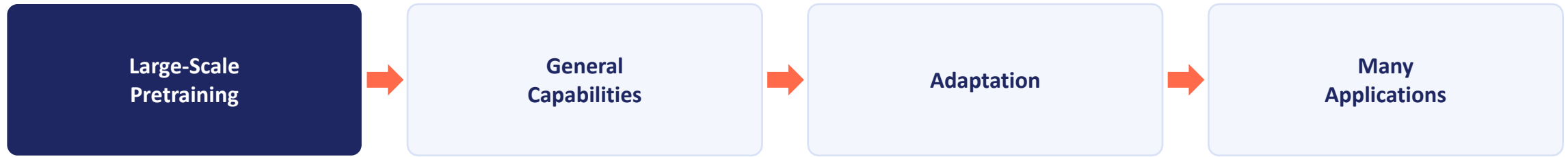
Generative AI



"Which Generative AI tools have you already used?"

Think-Pair-Share for 2 minutes, then share with the class.

Foundation Models

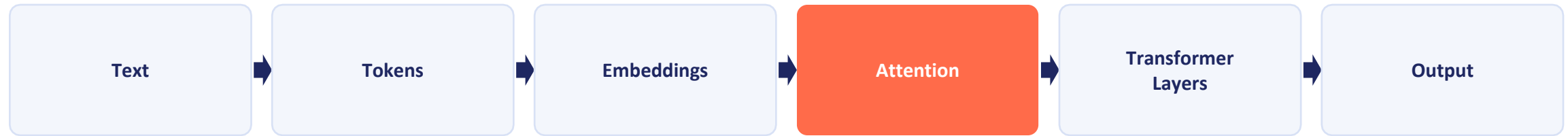


"Train once at large scale → adapt for many tasks."



(No training mathematics today — just the concept: general-purpose models adapted to specific tasks.)

Transformers: The Technology Behind Modern AI



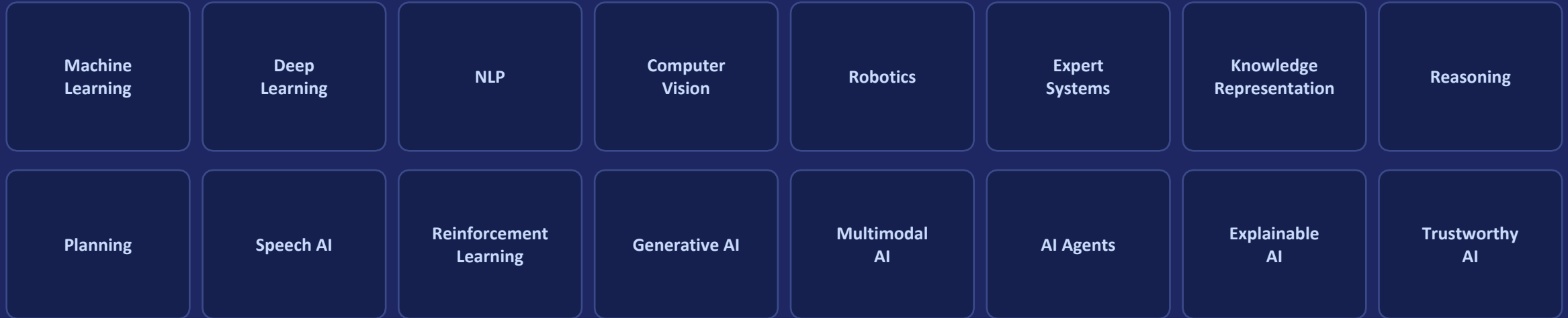
Attention, in plain language:

"The animal didn't cross the road because it was tired."

Attention lets the model weigh EVERY word against every other word to figure out that "it" refers to "the animal," not "the road."

No complex equations today — just the idea that context shapes meaning, word by word.

Major Branches of AI



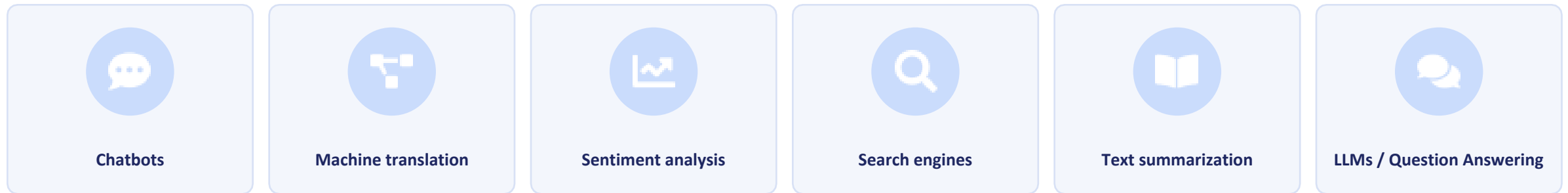
Computer Vision

Enabling computers to interpret and understand visual information.



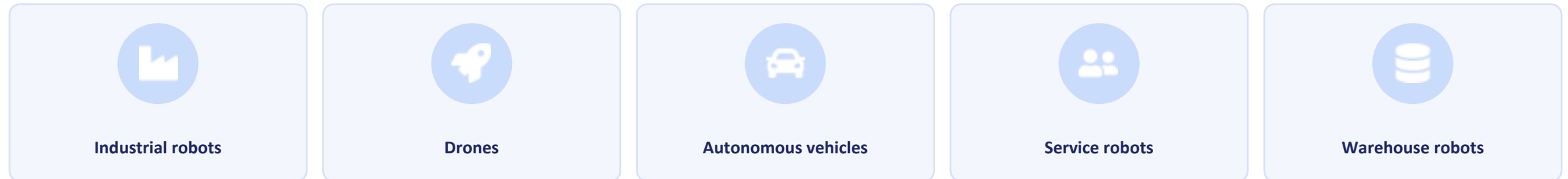
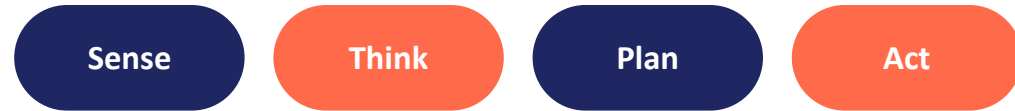
Natural Language Processing

Enabling computers to process, understand, and generate human language.



Robotics & Intelligent Agents

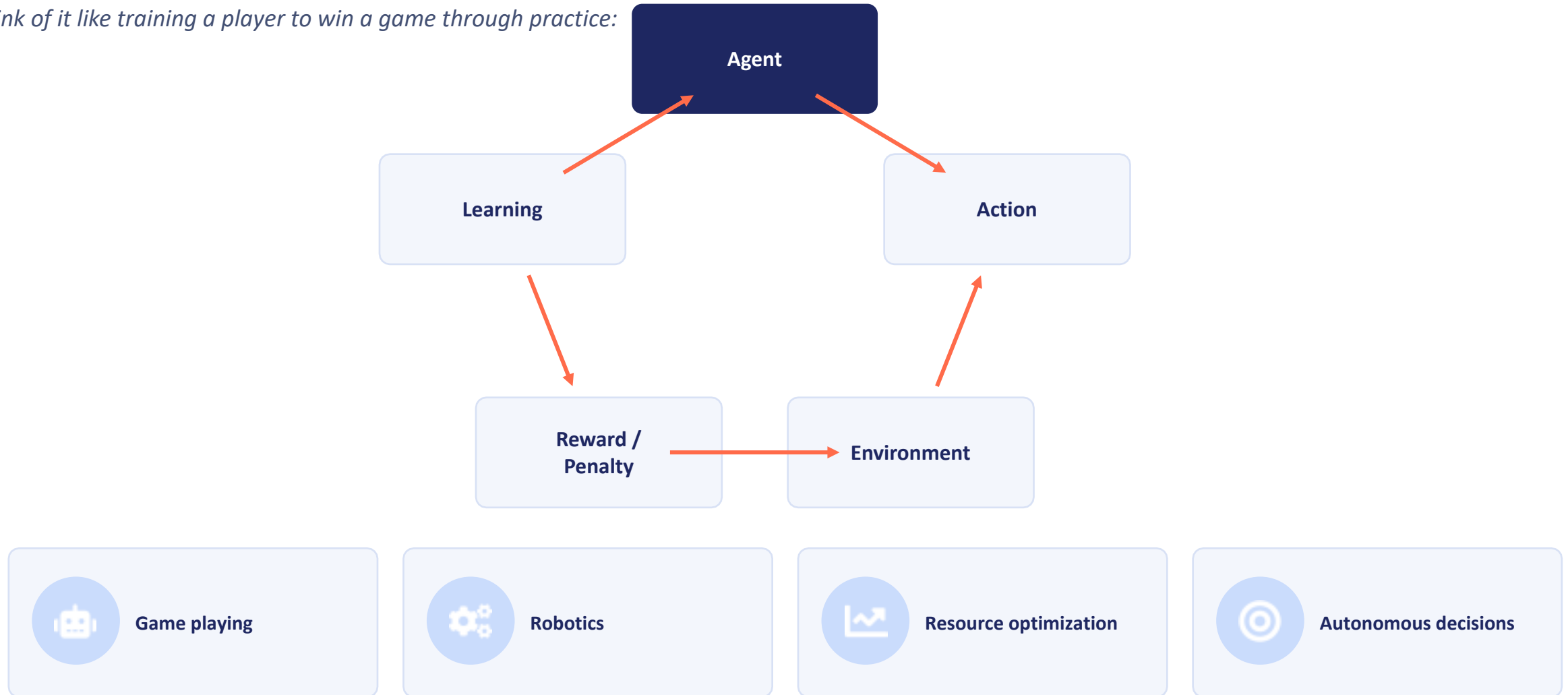
An AI agent must:



An AI Agent (basic idea): a system that perceives its environment, reasons/plans, and takes actions toward a goal.

Reinforcement Learning

Think of it like training a player to win a game through practice:



AI in the Real World



Healthcare

Medical image analysis



Finance

Fraud & risk detection



Education

Personalized tutoring



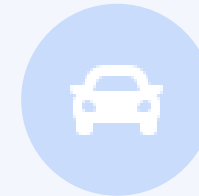
Cybersecurity

Threat detection



Agriculture

Crop monitoring



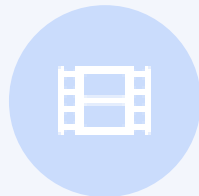
Transportation

Route optimization



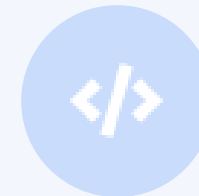
Manufacturing

Predictive maintenance



Entertainment

Content recommendation



Software Engineering

AI code assistance

AI in Computer Science



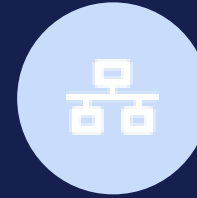
AI + Cybersecurity

Intrusion & anomaly detection



AI + Software Engineering

Code generation & review



AI + Networks

Traffic anomaly prediction



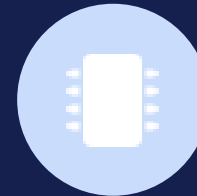
AI + Databases

Query optimization



AI + Cloud Computing

Auto-scaling & resource mgmt



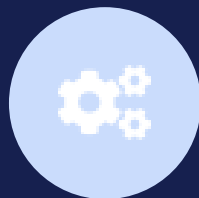
AI + IoT

Smart sensor decision-making



AI + HCI

Adaptive user interfaces



AI + Robotics

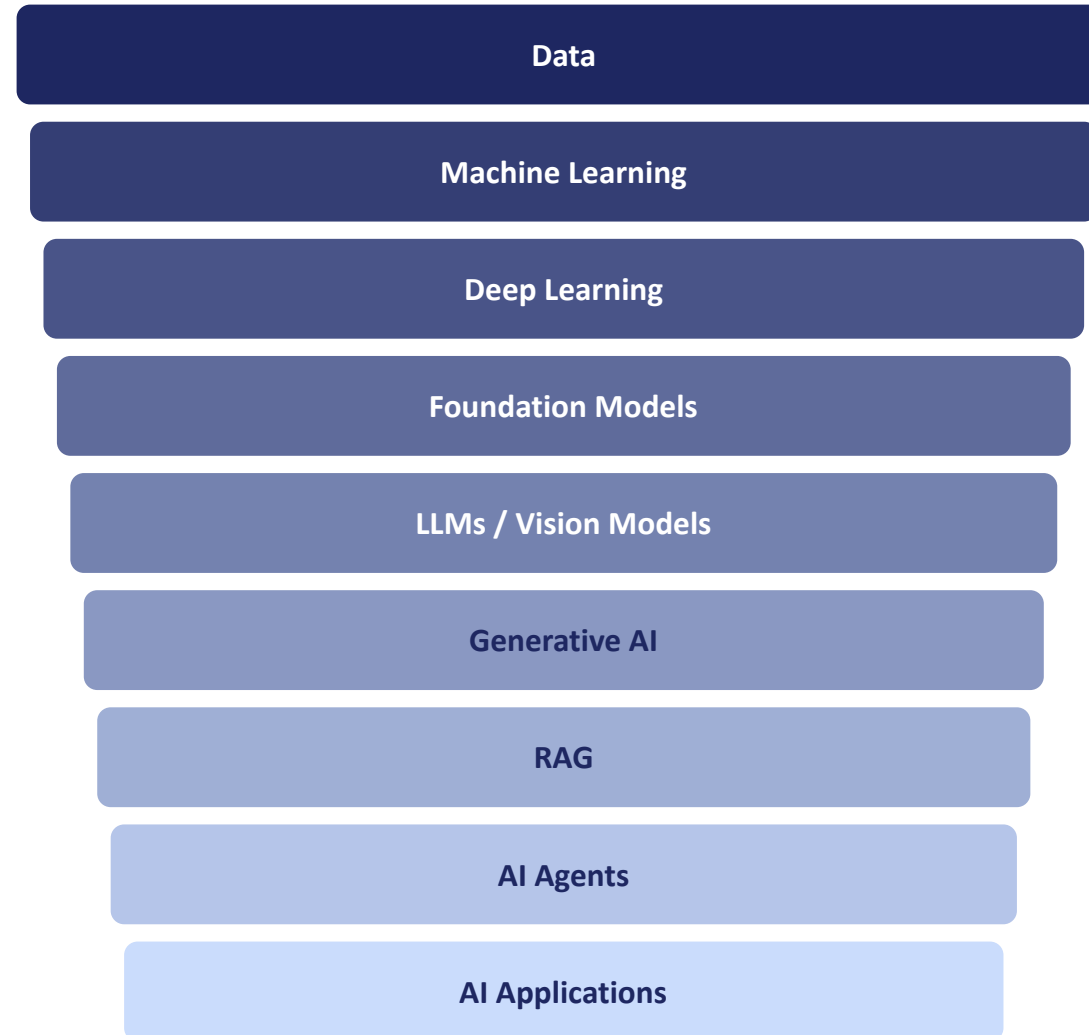
Autonomous control systems



AI + Data Science

Predictive analytics

Modern AI Ecosystem



Students will meet these technologies throughout the BS degree and in industry roles.

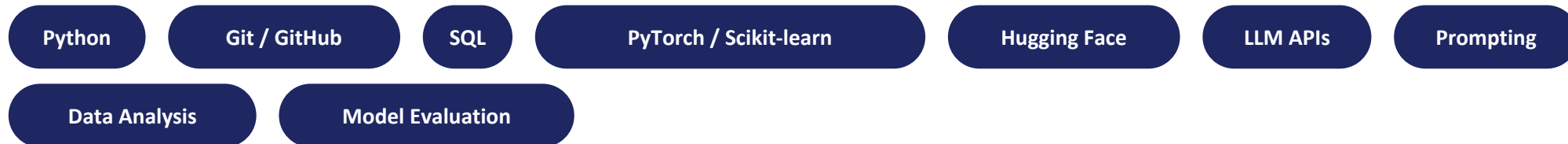
**Today: just the map.
Later courses: the depth.**

AI Skills for the Future

What Should a Computer Science Student Learn?



Tools & skills you'll build hands-on experience with:



This is a learning roadmap for your whole degree — not something to master all at once.

Can You Design an AI Solution?



Group 1

Detect fake news



Group 2

Detect cyber attacks



Group 3

Predict student performance



Group 4

Recognize plant diseases



Group 5

Build a university chatbot



Group 6

Detect road accidents

Each group answers:

1. What is the problem?
2. What data might be needed?
3. What AI capability is required?
4. Which AI branch could solve it?
5. What could be the benefit?

**5–7 minutes
to discuss**

30–60 sec
per group to present

From AI User to AI Problem Solver



"AI is not just about ChatGPT or robots."

"AI is a way of building computational systems that can learn, reason, perceive, generate, and make decisions."

"What real-world problem would YOU like to solve using AI?"

Your AI journey starts today.